

Problem Statement

kk-payments currently deploys without an isolated staging environment, making it difficult to validate changes independently before production. Environment-specific configuration is also mixed with application deployment configuration, while deployment validation and health monitoring require stronger automation. The capstone will provision a kijani-staging environment using Terraform and Ansible, update the Jenkins pipeline to deploy and validate staging before the production approval gate, separate staging configuration from the application image, and integrate Prometheus monitoring and the Week 10 serverless receipt chain.

Track

Track A : Infrastructure-First

This track focuses on multi-environment infrastructure, automated delivery, Kubernetes reliability, and observability.

Track A - Terraform + Ansible + Jenkins + Kubernetes + Prometheus + required serverless receipt integration.

In-Scope Components

- **Infrastructure:** Terraform provisions the kijani-staging environment reproducibly and maintains its infrastructure state.
- **Configuration:** Ansible manages server configuration, security, systemd, and logging.
- **CI/CD:** Jenkins deploys the application to staging, executes a smoke test, and provides a production approval gate.
- **Application platform:** Containerized kk-payments runs on Kubernetes with health checks, resource controls, and automatic recovery.
- **Observability:** Prometheus monitoring and alerting for kk-payments.

Out-of-Scope

- **Application feature development:** Business functionality is not changed because the capstone focuses on infrastructure, delivery automation, reliability, and observability rather than application features.
- **Kubernetes cluster creation:** A new Kubernetes cluster is not provisioned because the capstone uses the existing cluster and focuses on creating an isolated staging namespace within it.

Success Criteria

1. A merge to main triggers staging deployment automatically, and the staging smoke test passes before the production approval gate is presented.
2. The kijani-staging environment is reproducibly provisioned and configured, with kk-payments using environment-specific configuration and Kubernetes health checks, resource controls, and multiple Pods.
3. A defined kk-payments health condition triggers the Prometheus alert, and a staging receipt event successfully reaches kk-payments-receipts-staging and triggers the Week 10 serverless receipt chain.